

Testing Document – KUCampus Bite Project

1. Introduction

The KUCampus Bite project is a campus-based food ordering and delivery system designed to streamline the process of ordering meals from university canteens. The system provides interfaces for students and administrators to manage menu listings, place orders. This testing document outlines the testing process, types, and outcomes carried out to ensure system reliability, accuracy, and usability.

2. Object of Testing

The objective of testing KUCampus Bite is to ensure that the system functions according to its defined requirements. Testing aims to detect any software defects, verify that all modules integrate seamlessly, and confirm the system's performance and usability across different devices and network conditions.

3. Scope of Testing

Testing covers functional and non-functional aspects of KUCampus Bite, including login, menu browsing, order placement, and admin operations such as menu management and report generation.

4. Type of Testing

- a. Unit Testing: Each module such as login, menu display, and order placement was tested individually to ensure proper functionality.
- b. Integration Testing: Integration between student, admin, and canteen modules was tested to verify data flow and communication.
- c. System Testing: The complete system was tested to evaluate overall behavior under typical usage scenarios.
- d. Regression Testing: Re-testing was performed after bug fixes and updates to ensure no new issues were introduced.
- e. User Acceptance Testing (UAT): Final testing conducted by the Cantech team and their friends to confirm system readiness for deployment.

5. Test Scenarios

Scenario ID	Description	Expected Outcome
TS-01	Verify student login functionality	Valid users can log in successfully
TS-02	Check menu visibility for selected canteen	Menu displays all items correctly
TS-03	Test order placement flow	Order confirmed and stored in database
TS-04	Test admin adding new menu item	Item appears to students instantly
TS-05	Check response time during peak usage	System responds within 3 seconds

6. Test Cases

Test Case ID	Test Scenario	Test Steps	Expected Result	Status
TC-01	Login Functionality	Enter valid credentials and click login	Dashboard loads successfully	Pass
TC-02	Menu Display	Select a canteen and open menu	Menu items displayed accurately	Pass
TC-03	Order Placement	Add items to cart and confirm order	Order confirmed message shown	Pass
TC-04	Admin Menu Update	Admin adds a new item	Item visible to users immediately	Pass
TC-05	Response Time	Perform order placement	Response within 3 seconds	Pass

7. Bugs and Issues

- UI alignment issue on smaller mobile screens.
- Minor inconsistency in admin report timestamp format.

All reported issues were resolved in subsequent builds.

8. Tools Used

- Frontend Testing: Google Chrome DevTools
- Backend Testing: Postman for API testing, MongoDB Query Analyzer
- Version Control: GitHub

9. Test Summary

A total of 20 test cases were executed. Out of these, 18 passed successfully, and 2 minor UI-related bugs were identified and fixed. The application meets the functional and performance criteria as defined in the project requirements. The testing confirmed system readiness for real-world campus deployment.

10. Conclusion

The KUCampus Bite system successfully passed all critical test cases. The testing process validated that the application delivers a smooth, efficient, and reliable user experience for both students and canteen staff. The system is now considered ready for deployment and live use within the campus environment.